



SpiderRock V7 Tool Concept Guide

Order Entry Tickets

via Symbol Viewer

October 11, 2022

STOCK & FUTURE QUICK TICKET

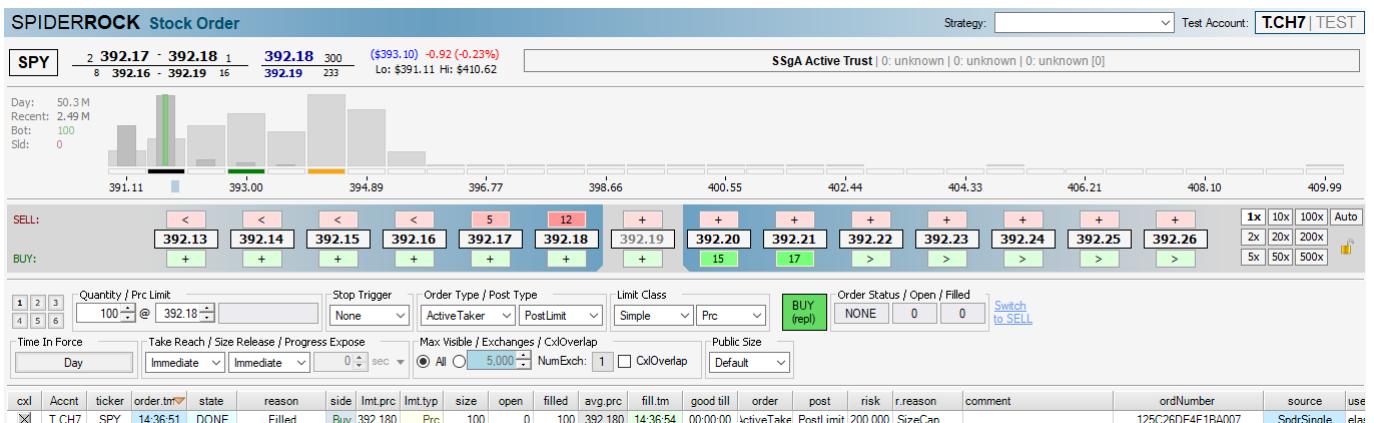
- The stock and future Quick Tickets allow users to send stock and future single orders directly **from the Symbol Viewer**.
- Double-clicking on the stock bid/ask prices in the upper left section of the RiskView tab, launches a stock quick ticket. In addition to the best price, we display the second price level with corresponding size. To the right, we also display the last two prints with corresponding size, the closing price, move from closing price in dollars and percentage as well as the high and low price levels on the day.

SPIDERROCK Symbol Viewer		TEST ACCOUNT	
SPY	10 350.38 - 350.39 89	350.39 200	(\$348.33) +2.06 (+0.59%) Lo: \$348.16 Hi: \$350.46
	22 350.37 - 350.40 38	350.38 100	

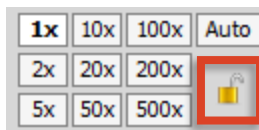
- Double-clicking on the futures bid/ask prices in the upper left section of the RiskView tab below the spot price launches a futures quick ticket.

SPIDERROCK Symbol Viewer	
ESU0	104 3,503.50 - 3,503.75 105 3,503.50 1 (\$3,485.30) +18.20 (+0.52%) Lo: 3,480.75 Hi: 3,509.50
	151 3,503.25 - 3,504.00 164 3,503.75 2
0 ESU0 +0 +0.00% 3,503.50 - 3,503.75 ESZ0 +0 +0.00% 3,493.00 - 3,493.25 ESH1 +0 +0.00% 3,483.25 - 3,485.00 ESM1 +0 +0.00% 3,457.00 - 3,480.75 ESU1 +0 +0.00% 3,438 - 3,497	

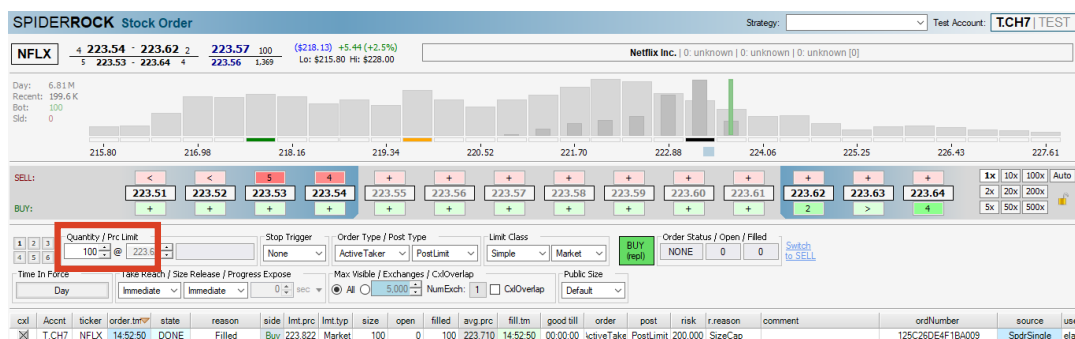
- Both the stock and futures order entry tickets display a print histogram of trades and corresponding prices done during the trading session (see screenshot below). The legend for the histogram is as follows:
 - Light gray bars:** Print distribution by price level during the trading session
 - Dark gray bars:** Weighted print distribution by most recent activity (10 minute coefficient)
 - Green bar:** official exchange closing mark (previous day)
 - Orange line:** first print price of the day during regular market hours
 - Black line:** the last print
 - Gray line:** the previous print
 - Light gray line:** the print before the previous
 - Red bar:** Client's last sell order
 - Green bar:** Client's last buy order
 - Some statistics appear to the left of the histogram providing the volume on the day, recent volume traded in the past 10 minutes and bought and sold client activity.



- Price ladder: The ladder is a visual representation of the current market depth displaying prices and sizes. The top row displays bids (red) and the bottom row shows offers (green).
- For the stock ticket, there is a 100 multiplier associated with the size displayed, where 1 represents 100 shares. For the future ticket, each number represents 1 future contract.
- The user can choose whether they want the ladder to visually move or be locked as the market ticks. Users can also choose the price scope by clicking on 1X, 2X, 5X, 10X, 20X, 50X, 100X, 200X, or 500X boxes.



- “Locked Mode” will display gray bars outside of the price spread. The ladder is always centered around the mid-market. There is a visual cue (blue frame around the price box) that follows the anchor price, which is the initial mid-market. If the anchor price moves out of the ladder range, it is re-assigned to the current mid-market.
- “Unlocked Mode” will display blue bars outside of the price spread. The ladder prices are fixed and the blue bars representing the bid/ask spread are moving. If the current market moves out of the ladder range, the ladder is re-centered.
- The client can select any order parameters using the order entry section below the price ladder. Orders can be launched using the sell/buy buttons or dynamically by using the price ladder. For more information on order parameters, please refer to the Ticket Blotter section.
- Once the quantity and order parameters are selected, the client can use the ladder to quickly launch an order by ctrl + left clicking on any of the buy/sell price boxes of the ladder (above and below the prices). The top line creates a sell order and the bottom line creates a buy order. The number of shares will be displayed in the box that was clicked (1 represents 100 shares). A new working order will also be visible in the Order grid below the order parameters section.



- The order grid is a convenient order management view and can be configurable by right clicking on the headers. For greater details of working and filled orders, the client should refer to the **Order Manager** tool.

OPTION QUICK TICKET

The Option Quick Ticket allows users to send and monitor single-strike option orders directly from the Symbol Viewer. Double clicking on any bid/ask price or implied vols from the option montage section of the RiskView tab will launch a quick ticket and allow a user to set a number of order entry parameters to execute an order.

SPIDERROCK Option Order Strategy: [dropdown] Test Account: TCH7 | TEST

GOOGL 3 101.73 - 101.74 14 101.74 100 (\$102.91 -1.17 (-1.1%))
6 101.72 - 101.75 1 101.74 121 Lo: \$100.93 Hi: \$107.85

Alphabet Inc | 0: unknown | 0: unknown | 0: unknown [0]

call market
(NOBX) 65 2.75 - 2.80 24 (NOBX) | 2.84/A 1 (BOX,1min) | 36.38 - 48.71 | **SR: 2.77 48.71**

put market
(EMLD) 9,288 0.03 - 0.04 1,116 (C2) | 0.04/ 1 (BATS,1min) | 41.21 - 43.69 | SR: 0.04 43.69

GOOGL / SEP-16 2022 / \$99 C uPrc: \$101.735, de: +0.947 valid range: \$96.65 - \$106.82 1.0% 2.0% 5.0% sdiv: 0.00%, ddiv: \$0.00 Exchanges (All)

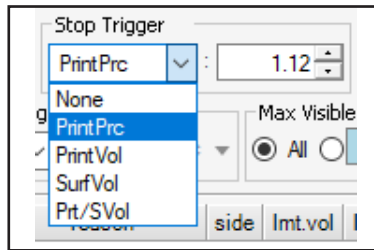
Quantity / Prc Limit: 1 @ 2.76 39.81% Stop Trigger: None Order Type / Post Type: ActiveTake / NoPost Limit Class: Simple Prc: Ref SDiv: Locked Switch to BUY

Auto Hedge: SM NORM Time In Force: Day Take Reach / Size Release / Progress Expose: Immediate Immediate 0 sec Max Visible / Exchanges / CxlOverlap: All 5,000 NumExch: 1 CxlOverlap: Opx Floor: 0.00 Public Size: Default

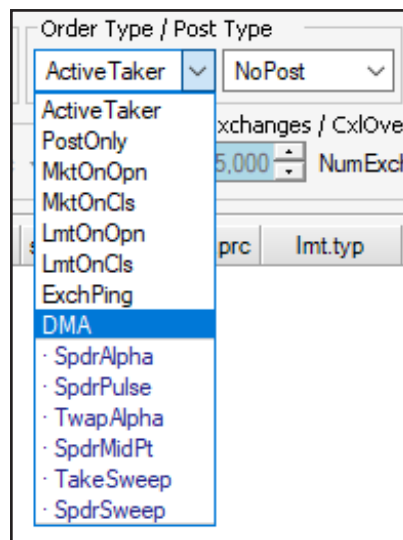
cxl	acct	root	expire	xx	cp	order.tr	state	reason	side	lmt.vol	lmt.prc	lmt.typ	size	open	filled	avg.prc	avg.vol	fill tm	good till	order	post	auto	risk	r.reason	comment
X	T.CH7	GOOGL	2022-09-16	99.00	C	11:28:29	DONE	Filled	Sell	0.00	2.76	Prc	1	0	1	2.760	36.5	11:28:37	00:00:00	ActiveTaker	None	SmartNorm	656	MinGrpDD	

- Similar to the stock and future Quick Tickets, the upper left corner of the option Quick Ticket displays the underlying symbol, the top two price and size levels, the last two prints, previous closing price, dollar and percentage day move from the closing price, as well as high and low price levels on the day.
- The order parameters section is tailored to trading options. The boxes directly below the general price information reflects the call and put markets. These allow the user to click on any implied volatilities or prices to automatically populate that value as a volatility or price limit. In addition, the SpiderRock surface price and volatility surface is displayed (as shown in the red rectangle above).
- Parameters to enter orders are displayed under the general price information. They are defined by the user to guide the execution behavior of an order. For more information on these parameters, please refer to the **Ticket Blotter section**.
- Features that are unique to the Option Quick Ticket:

Valid range allows for a user to set specific min and max underlying price ranges. The option order will only be working if the underlying price is within the min/max range.



Stop Trigger allows a user to set either a price/vol level that will trigger the order to be sent to the market. PrintPrc will trigger off of a print price, PrintVol will trigger off of the vol level of a print, SurfVol will trigger when the SpiderRock surface reaches that vol limit, Prt/SVol will trigger when either a vol level of a print or the SpiderRock surface is reached.



DMA allows for users to send Direct Market Access price orders to a single exchange. These orders will not route away from the initial exchange. By default SpiderRock will choose the specific exchange unless the user overrides.

- All orders sent in the quick ticket appear below the order parameter section. This window provides individual working orders and displays order details such as current state, fill, amount open, average price, time remaining and any potential risk limits that the order may hit.
- To restore an order that was previously sent from the quick ticket, double click on that order row. This will populate the ticket with those order parameters. The order tabs in the **Order Manager** tool provide additional functionalities to manage working orders.

TICKET BLOTTER

Option Ticket

The Ticket Blotter gives users more flexibility to trade a strip of options across multiple strikes, expiries, multiple symbols and also spread orders (multi-leg orders).

The screenshot shows the SPIDERROCK Ticket Blotter interface. The 'Option' tab is selected. The interface includes various input fields for order parameters, a table of orders, and sections for limits and today's orders. A red arrow points to the radio buttons (VOL, PRC, TKT) and a green arrow points to the 'Option' tab. A blue arrow points to the 'Today' tab in the 'TODAY'S ORDERS' section.

- Radio buttons (red arrow above):

The 6 radio buttons to the left of the ticket can be used to save all selected algo parameters by clicking on the Shift key and 1 out of the 6 available buttons. Clicking on one of these pre recorded buttons will reload the selected algo parameters and streamline the order entry process.

- Ticket type (green arrow above):

VOL: Volatility order (Option mode only)

PRC: Price order

TKT: For entry of away tickets

- Order View tabs (blue arrow above):

Active: Working orders generated by the ticket blotter

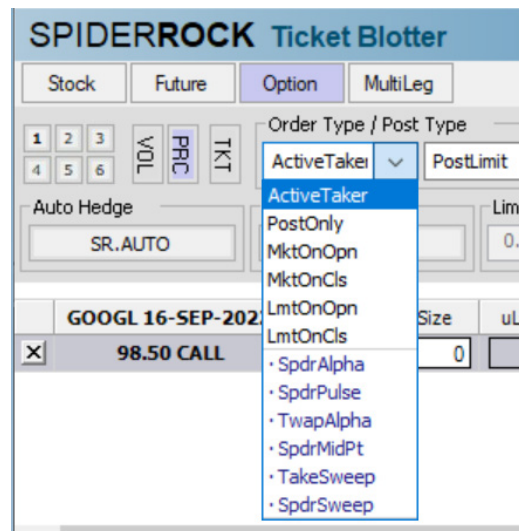
Today: All working, filled, or canceled orders during the current trading session

History: Ticket Blotter order history from the previous 5 days of trading

Staged: Orders staged in the system via FIX or SRSE (SQL API) appears under the staged tab

LmtOnCls: Send a limit order on close





- Order Type:

ActiveTaker: Orders sent to the market will remove liquidity only

Postonly: Orders sent to the market will post liquidity only

MktOnOpn: Send a marketable limit order on open

MktOnCls: Send a marketable limit order on close

LmtOnOpn: Send a limit order on open

LmtOnCls: Send a limit order on close

Pre-defined order type:

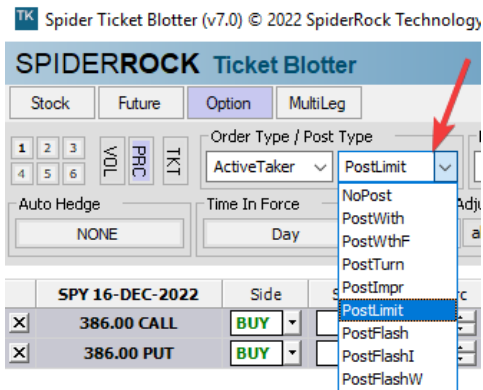
SpdrAlpha, SpdrPulse, and TWAPAlpha all work parent orders in distinct modes at increasing levels of aggressiveness. These orders can be used with or without a parent limit and with or without any secondary limits.

SpdrAlpha: Utilizes one slice which starts in passive mode at and moves through the moderate and aggressive modes until crossing the market at the end of the time duration. The full size of the order is available for the entire time period.

SpdrPulse: SpdrPulse is designed to work each parent order in distinct slices. Each slice will work through passive, moderate, and aggressive modes. If any slice fills, the order will immediately revert to passive mode and start working the next slice. The SpdrPulse is designed to work each parent order in distinct slices duration for each SpdrPulse slice is 30 seconds but can be set on each individual parent order. SpdrPulse is available for equities, futures, options, and MLEG orders.



TwapAlpha:	This progression algo that is intended to work for the full duration of the parent order. This algo determines a slice size, count, and target duration. It then works the next slice through the different modes (passive, moderate, and aggressive). If a slice fills before its target time has elapsed, the algo waits until the fully allotted time has passed, before switching to the next slice. TwapAlpha is available for equities, options, futures, and MLEG orders. There is also a variation for equities called TwapAlphaC that will promote any parent order balance into the primary equity closing auction just before auction cutoff time. SRSE and FIX users can define their own participation level in the closing auction by populating the Vwap participation field in the order gateway or tag 5058, via FIX.
SpdrMidPt:	Order is pegged on the making and taking side at a 50% alpha probability
TakeSweep:	Allows users to sweep the market using ISO orders (available on account configuration)
SpdrSweep:	Send a marketable limit order designed to sweep existing liquidity across exchanges



Post Type: defines the posting behavior of the orders

- NoPost:** No post type selected.
- PostWith:** Join existing markets if there is sufficient limit room to do so. Will not improve an existing market. Will fade when alone and will be dynamically sized if child order would outsize the prevailing market size.
- PostWithF:** Join existing markets if the size on the join side is at least 25% of the opposite side. Will fade if the ratio between the sides drops below 20%. Designed to prevent your order from being the last at a price. Will use dynamic market sizing to prevent out sizing the prevailing market size.
- PostTurn:** Join or improve existing markets if there is sufficient limit room to do so. Will not improve own order unless joined with sufficient size. Will be dynamically sized if the child order would outsize the prevailing market size.
- PostImpr:** Join or improve an existing market if there is sufficient limit room to do so. Will improve own market periodically (~30 – 60 seconds) if there is sufficient limit room. Will be dynamically sized if the child order would outsize the prevailing market size.
- PostLimit:** Immediately post a public order at the limit price. Will improve or fade as limit price moves. Will be dynamically sized if the child order would outsize the prevailing market size.
- PostFlash:** Same as post limit, but flashes the order in the market for a 1 second and then is removed from the market. It will only flash if the child order is improving the market. It will flash again at a random time between 45 seconds.
- PostFlashI:** Similar to PostFlash. However instead of posting immediately at on the limit price, the order will first flash one price increment better than the NBBO. With each subsequent flash the algo will further improve. This will continue until the algo has reach the user limit. Like PostFlash, the order will post for 1 second and then go dark (no child orders) for a random time of 45 seconds. At the end of the timer the order will flash the market again.
- PostFlashW:** This is as PostFlash, where the order will flash in the market for 1 second, but will post on the bid for the remaining time.

CxlOverlap / Max Visible / Exchanges / Public Size

☐ CxlOverlap 100 % all NumExch: 1 PublicSize: Default

Child Orders	iBid	oBid	oAsk
	66.57	7.60	7.70

Max Visible (green arrow):

The percentage box represents the maximum percentage of the order size that will be visible in the market.

The NumExch is the maximum number of exchanges a user can select to post an order simultaneously. The maximum is set at 4.

NumExch (yellow arrow):

The user can post liquidity up on to 4 exchanges simultaneously using the NumExch box. The minimum available postage order size will be split evenly across the number of exchanges.

Public Size (blue arrow):

As a function of posting liquidity, the user can define a size mode to dictate the size being posted.

- Default:** Will use the public size options that has been assigned to the posting type selected. All posting algos default to market size except Post Limit and Post Flash which use FullSize.
- Randomize:** Will randomize the last 10% of the average market size.
- AvgMkt:** Uses dynamic sizing, which is the average of the public size on the NBBO and the second best price.
- FullSize:** References the user's full order size.
- FullSizeR:** Will randomize the last 10% of the user's full order size.

CxlOverlap (red arrow):

When checked, it allows the system to send another child order before receiving a cancel acknowledgment message. This may cause an overfill condition, particularly when working a take/make algo style. The default is to NOT have this box checked.

The screenshot shows the SPIDERROCK Ticket Blotter interface. At the top, there's a header with 'SPIDERROCK Ticket Blotter' and a 'Strategy' dropdown. Below this are tabs for 'Stock', 'Future', 'Option', and 'MultiLeg'. The 'Option' tab is selected. The interface includes various input fields for order entry, such as 'Order Type / Post Type', 'ActiveTake', 'PostLimit', 'Limit Class' (set to 'Simple'), 'Market', 'Exchanges', 'Ref SDiv', 'Locked', 'Send New', 'Update Existing', 'Take Reach / Size Release / Progress Expose', 'CxlOverlap / Max Visible / Exchanges / Public Size', 'CxlOverlap', 'NumExch', 'PublicSize', 'Opx Floor', and 'UPrc Range (%)'. A table of open orders is displayed at the bottom, with columns for 'GOOGL 16-SEP-2022', 'Side', 'Size', 'uLimitPrc', 'Open', 'Filled', 'State', 'Limit', 'Child Orders', 'CumBSz', 'oBid', 'oAsk', 'CumASz', 'IBid', 'IAsk', 'uBid', 'uAsk', 'sVol', 'sPrc', and 'optVe'. The table shows several orders for 'GOOGL 16-SEP-2022' with 'BUY' side, including '101.00 CALL', '101.25 CALL', '101.50 CALL', '101.75 CALL', and '102.00 CALL'.

- Selecting a limit class:

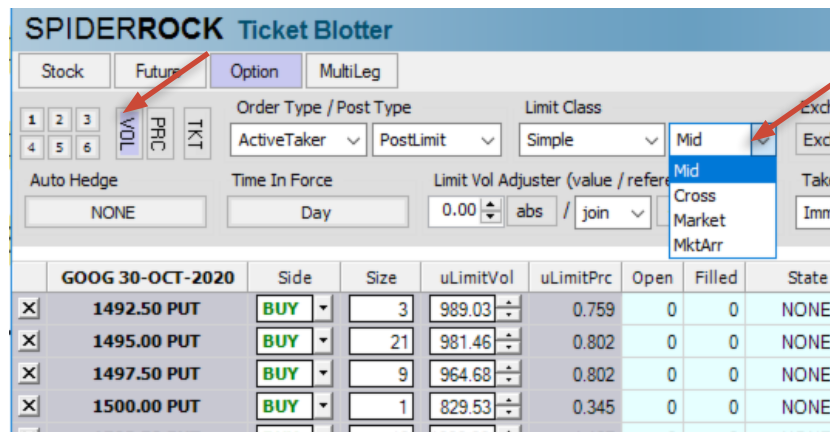
SpiderRock parent orders carry both a primary limit and (optionally) a secondary limit. If both primary and secondary limits have been established, then the effective limit at which the order will be represented in the market will be the most restrictive of the primary and secondary limits.

Primary limits can either be static (simple vol/price limits). Secondary limits are expressed as an offset from the SpiderRock volatility surface for options only or alpha probabilities. The secondary limit will often be the most restrictive while the primary limit will function as a top/bottom limit to prevent execution outside of a volatility or price level.

Available Limit Classes are:

- Simple:** User defined primary limit (vol/price)
- Surface:** Secondary limit expressed as an offset of the SpiderRock volatility surface
- Probability:** Secondary limit expressed as an alpha probability
- mkSurf/tkProb:** Allows users to leverage the SpiderRock volatility surface on the making side and the alpha probabilities on the taking side as secondary limits





- If the user selects a vol based limit (VOL), the user has the following choices to set the effective limit:

Mid: The vol order uses the midpoint of the underlying as the reference point for the vol to price conversion.

Cross: Vol order uses the bid/ask price of the underlying as the reference point for the vol to price conversion.

This causes the vol to price conversion to be more conservative than using the midpoint. This is order directional.

Market: Price increment through the market; resets when the NBBO price changes

MarketArr: Sets a limit price .10 through the market based on market arrival price. The price does not reset.



SPIDERROCK Ticket Blotter

Stock Future **Option** MultiLeg

1 2 3 VOL **Prc** TKT
4 5 6

Order Type / Post Type: ActiveTaker PostLimit Limit Class: Simple Prc

Auto Hedge: NONE Time In Force: 4:00 pm Limit Prc Adjuster (value / reference): 0.00 abs / surf +bid

	RACE 12-JUN-2020	Side	Size	uLimitPrc	Open	Filled
X	149.00 CALL	BUY	45	24.80	37	8
X	150.00 CALL	BUY	1	23.80	0	1
X	152.50 CALL	BUY	36	21.20	36	0
X	155.00 CALL	BUY	48	18.30	48	0
X	157.50 CALL	BUY	74	16.30	74	0

Limit Class dropdown menu options: Prc, PrcV, PrcVX, PrcDe, PrcDeX, PrcDeT, PrcDeP, PrcDeXT, PrcDeXP, RelMid, RelCrx, RelTurn, Market, MktArr

- If the user selects a price based limit (PRC), the following choices are available for the effective limit:

Prc:	Static limit price; does not change with underlying symbol price
PrcV:	Limit price is converted into a Vol limit using the mid-point of the underlying market as a reference price
PrcVX:	Limit price is converted into a Vol limit using the using bid/ask (hedge direction) of the underlying market as a reference price
PrcDe:	Effective limit = initial price + (de * dUPrc) + (0.5 *ga * dUPrc * dUPrc) based off of the mid-point of the underlying. dUPrc being the change in the UP
PrcDeX:	Same as PrcDe but uses underlier bid/ask (hedge direction) instead of mid-point
PrcDeT:	Same as PrcDe but include time decay
PrcDeP:	Same as PrcDe but price is protected
PrcDeXT:	Same as PrcDeX but includes time decay
PrcDeXP:	Same as PrcDeX but price is protected
RelMid:	This limit works in conjunction with the size release Immediate: Limit will be put at mid-market and won't change Fast reset: Limit will be put at mid-market and reset on and off every 15 seconds Slow reset: Limit will be put at mid-market and reset on and off every minute
ReCrx:	Same as above, but limit will cross the market
ReTurn:	Same as above, but limit will turn the market by one price increment
Market:	Sets a limit price 1 price increment through the market; resets when the NBBO price changes
MarketArr:	Sets a limit price .10 through the market based on market arrival price. The price does not reset

	SPY 28-AUG-2019	Side	Size	uLimitPrc	Filled	Stat
X	292.50 PUT	BUY	200	5.99	0	NOM
X	293.00 PUT	BUY	200	6.48	0	NOM
X	293.50 PUT	BUY	50	6.98	0	NOM
X	294.00 PUT	BUY	200	7.47	0	NOM

- User-defined primary limits (Prc/Vol) can be easily set using the Limit (Prc/Vol) Adjuster. User limits can be set based on different market reference points. The reference points are:

- Surf:** The user-defined limit references the prevailing SpiderRock volatility surface
- Join:** The user-defined limit references joining the market by calibrating on bid price for buy orders and ask price for sell orders
- Mid:** The user-defined limit references the prevailing mid-mark
- Theo:** The user-defined limit references the user theoretical values
- B/S:** The user-defined limit references the user buy/sell levels
- Adj:** The user can make adjustments using the value selector to existing limit for working orders
- Cross:** The user-defined limit references crossing the market by calibrating on ask price for buy orders and bid price for sell orders

User can further offset from these reference points by using the up and down arrows of the value selector. A positive number will be "more aggressive than" and a negative number will be "less aggressive than".

SPIDERROCK Ticket Blotter

Stock Future Option MultiLeg

Order Type / Post Type: ActiveTaker PostLimit Limit Class: Surface Mid Probability Limit: Static MakeLimit: 0.50 TakeLimit: 0.50

Auto Hedge: NONE Time In Force: 4:00 pm Limit Vol Adjuster (value / reference): 0.00 abs / surf +bid +ask Take Reach / Size Release: Immediate Immediate Max Visible: CxOverlap 100 % all NumExch: 1 PublicSize: Default

Surface Limit: PrcOffset: 0.010 VolOffset: 0.00 %

	SPY 28-AUG-2019	Side	Size	uLimitVol	uLimitPrc	surfPrc	effLmtPrc	Open	Filled	State	Limit	oBid	oAsk	iBid	iAsk	uBid
X	292.50 PUT	BUY	200	17.86	5.875	5.885	5.870	0	0	NONE		5.83	5.99	0.00	24.23	2
X	293.00 PUT	BUY	200	17.90	6.363	6.373	6.360	0	0	NONE		6.32	6.48	0.00	25.36	2
X	293.50 PUT	BUY	50	17.98	6.856	6.866	6.850	0	0	NONE		6.76	6.98	0.00	26.87	2
X	294.00 PUT	BUY	200	18.15	7.352	7.362	7.350	0	0	NONE		7.32	7.47	0.00	27.90	2

The user can define a secondary limit based on the SpiderRock volatility surface or the alpha probabilities.

If surface is chosen, the user can additionally provide a price/vol offset to the SpiderRock volatility surface using the Surface Limit box. A positive number will be a more aggressive offset from the surface. A negative number will be a less aggressive offset from the surface.

When selecting Surface as a secondary limit, the grid displays the following parent order limit information:

uLimitVol: User-defined primary limit in vol terms

uLimitPrc: Conversion of the uLimitVol in price terms

surfPrc: Secondary limit based on SpiderRock surface +/- offsets

effLmtPrc: The effective limit price is the tightest limit between the primary and secondary limits



SPIDERROCK Ticket Blotter

Stock Future **Option** MultiLeg

1 2 3 4 5 6 VOL PRC TKT

Order Type / Post Type: ActiveTaker PostLimit

Limit Class: Probability Mid

Probability Limit: Static MakeLimit: 0.50 TakeLimit: 0.50

Auto Hedge: NONE

Time In Force: 4:00 pm

Limit Vol Adjuster (value / reference): 0.00 abs / surf +bid +ask

Take Reach / Size Release: Immediate Immediate

Max Visible: ☐ CxlOverlap 100 % all

	SPY 28-AUG-2019	Side	Size	uLimitVol	uLimitPrc	mkProbPx	tkProbPx	mkEffLmtPx	tkEffLmtPx	Open	Filled	State
X	292.50 PUT	BUY	200	17.88	5.888	5.848	5.848	5.848	5.848	0	0	NONE
X	293.00 PUT	BUY	200	17.90	6.349	6.333	6.333	6.330	6.330	0	0	NONE

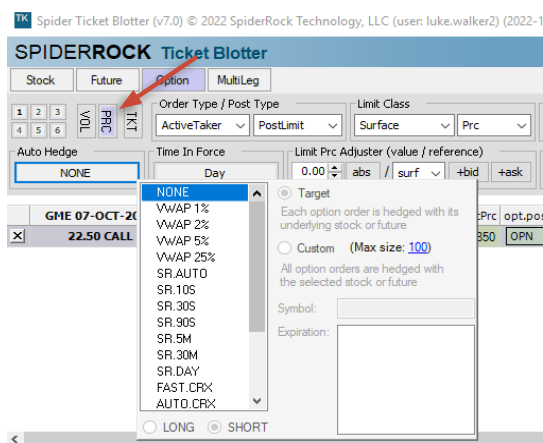
If the secondary Limit is **Probability**, users can offset the SpiderRock Probability using the Probability Limit box as follows:

- Static:** User-defined probability level implemented via the MakeLimit or TakeLimit boxes
- Relative:** User can select the average trailing 10-minute z score and can adjust the std deviation
- Eagle:** Pre-canned make/take probability limits set at 0.50 (make side) and 0.50 (take side)
- Hawk:** Pre-canned make/take probability limits set at 0.45 (make side) and 0.35 (take side)
- Falcon:** Pre-canned make/take probability limits set at 0.40 (make side) and 0.15 (take side)

When selecting Probability as a secondary limit, the grid displays the following parent order limit information:

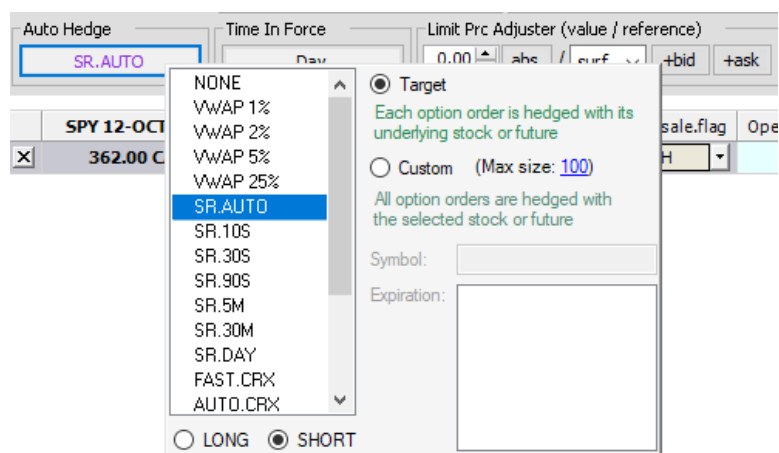
- uLimitVol:** User-defined primary limit in vol terms
- uLimitPrc:** Conversion of the uLimitVol in price terms
- mkProbPx:** Secondary limit based on make SR probabilities
- tkProbPx:** Secondary limit based on take SR probabilities
- mkEffLmtPx:** Making effective limit - most conservative limit between the primary & the secondary make limit (also effected by the posting algo the user has chosen)
- tkEffLmtPx:** Taking effective limit - most conservative limit between the primary & the secondary take limit



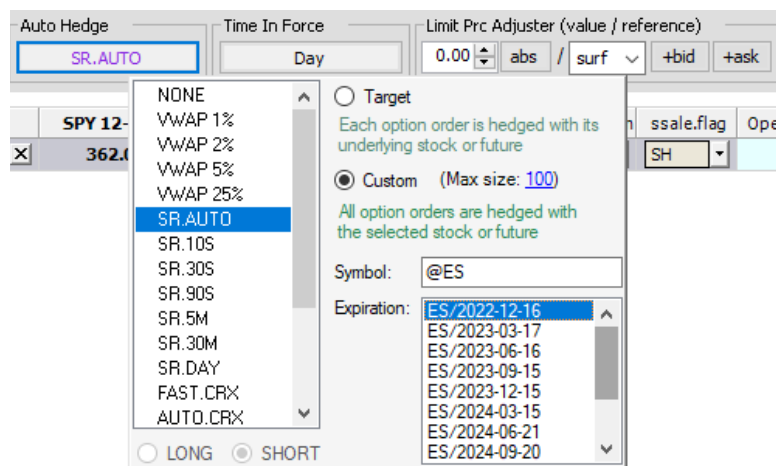


- Selecting an Auto Hedge strategy

- VWAP 1%:** The hedge will release additional size as the market trades. The available parent size will be 1% of market volume
- VWAP 2%:** The available parent size 2% of market volume
- VWAP 5%:** The available parent size will be 5% of market volume
- VWAP 25%:** The available parent size will be 25% of market volume
- SR AUTO:** smart TWAP – Automatically select a time duration based on liquidity profile of the underlier
Takes into consideration the volume of the underlying stock from the prior business day:
If StockVolume > 25MM then we will hedge in 4 slices with a 10 second duration.
If 10MM < StockVolume < 25MM then we will hedge in 10 slices with a 30 second duration.
If StockVolume < 10MM then we will hedge in 10 slices with a 90 second duration.
- SR 10S:** smart TWAP with a 10 second time duration
- SR 30S:** smart TWAP with a 30 second time duration
- SR 90S:** smart TWAP with a 90 second time duration
- SR 5M:** smart TWAP with a 5 minute time duration
- SR 30M:** smart TWAP with a 30 minute time duration
- AUTO-CRX:** order with a price that crosses the market one tick through (resets every 1s)
- FAST-CRX:** order with a price that crosses the market by one tick. If it misses the market, another order will be sent as soon as the cancel message is received; there is no reset time
- AUTO.MID:** order with a price that post at mid-market (resets every 20s)
- AUTO.TRN:** order with a price that betters the market by 1 price increment (resets every 45s)
- SM.FAST:** 75% of the bid/ask spread and becomes more aggressive following prints
- SM.NORM:** order with a price that posts mid-market of the bid/ask spread and becomes more aggressive following prints
- STATIC:** order with a price that posts at mid-market and does not reset



Target: Default setting - Option trades are hedged with the corresponding underlying instrument. If the underlying instrument is a cash instrument (e.g. SPX or VIX) we will hedge with the corresponding equity.



Custom: Users can click on the Custom button to change to a related instrument with which to hedge an option trade. Users can choose a stock or future symbol by typing in the Symbol window. If a future is selected, the user will then choose an expiration. For more information contact the Customer Support Desk: cs@spiderrock.net

SPIDERROCK Ticket Blotter

Stock Future **Option** MultiLeg

1 2 3 4 5 6 VOL PRC TKT

Order Type / Post Type: ActiveTake PostLimit Limit Class: Simple Market

Auto Hedge: SR.AUTO Time In Force: Day Limit Prc Adjuster (value / reference): 0.00 abs / cro +bid +ask

Type: ☒ Day ☐ ExtDay ☐ IOC

☐ Cut-off: 15:00:00 ☐ Duration: 0 sec (Times are EDT)

	GOOGL 16-SEP-2022	Side
X	101.00 CALL	BUY
X	101.25 CALL	BUY
X	101.50 CALL	BUY
X	101.75 CALL	BUY
X	102.00 CALL	BUY

482 0 482 DONE

- Selecting a Time In Force

The time in force is selected by the user and imposes the time that a particular order will work within the SpiderRock system, after which, the order would be canceled automatically if it has not filled. The choices are Day, Extended Day, IOC. In addition, the user can implement a specific cutoff time or duration in seconds.



Stock Future Option MultiLeg

Order Type / Post Type: ActiveTake PostLimit Limit Class: Simple Market Exchanges: Exchanges (All) Ref SDiv: Locked Send New

Auto Hedge: SR.AUTO Time In Force: Day Limit Prc Adjuster (value / reference): 0.00 abs / cro +bid +ask

Take Reach / Size Release / Progress Expose: Immediate Immediate 0 sec

	GOOGL 16-SEP-2022	Side	Size	uLimitPrc	Open	Filled	State	Limit	Child Orders	CumBS
X	101.00 CALL	BUY	369		60	309	WORK		[60 @ 0.94 N] +	257
X	101.25 CALL	BUY	437		19	418	WORK		[19 @ 0.74 N] +	257
X	101.50 CALL	BUY	222		0	222	DONE	0.410	[none]	127
X	101.75 CALL	BUY	885		0	885	DONE		[none]	3

- Take Reach / Size Release

Take reach applies to the taking side algo parameter. The choices are:

- Immediate:** The algo takes immediately if the limit is marketable
- Delayed:** The algo waits a few seconds before taking thus giving the posting side an opportunity to generate fills
- Passive:** The algo does not take until the market turns one price increment in the client's favor
- WeakOnly:** The algo only takes liquidity when the opposite side size is less than the average market size.
- FullSize:** This will send the full order size on the taking side of the algo
- ISO Sweep:** Utilizes ISO market code to sweep the market. Can only be used if setup to use ISO

Stock Future Option MultiLeg

Order Type / Post Type: ActiveTake PostLimit Limit Class: Simple Market Exchanges: Exchanges (All) Ref SDiv: Locked Send New

Auto Hedge: SR.AUTO Time In Force: Day Limit Prc Adjuster (value / reference): 0.00 abs / cro +bid +ask

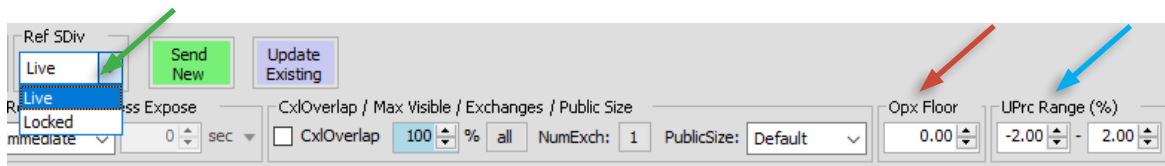
Take Reach / Size Release / Progress Expose: Immediate Immediate 0 sec

	GOOGL 16-SEP-2022	Side	Size	uLimitPrc	Open	Filled	State	Limit	Child Orders	CumBSz
X	101.00 CALL	BUY	369		0	369	DONE	0.960	TWAP.4	
X	101.25 CALL	BUY	437		0	437	DONE	0.760	TWAP.8	237
X	101.50 CALL	BUY	222		0	222	DONE	0.580	TWAP.12	237
									TWAP.Auto	485
									Slow Reset	

Parent orders can dynamically release order size over time using one of the following rules:

- Immediate:** Entire order size is immediately available
- TWAP.4:** The order working size is divided into 4 equal segments
- TWAP.8:** The order working size is divided into 8 equal segments
- TWAP.12:** The order working size is divided into 12 equal segments
- TWAP.Auto:** The system will select N number of segments to work the order according to stock liquidity
- Slow Reset:** Limit will be reset on and off every minute
- Fast Reset:** Limit will be reset on and off every 15 seconds





- Opx Floor [red arrow]

Users can implement an option price under which they would NOT want to buy or sell

- UPrc Range (%) [blue arrow]

This value defaults at up and down 2%. It represents the underlier price range allowed for an option order to work. If the underlying price moves out of set range, the option order is put on risk hold until the underlier price moves back into the set range.

- Ref SDiv [green arrow]

A locked sdiv will not change throughout the life of the parent order. A live sdiv will continually adjust in real time throughout the life of the parent order

SPIDERROCK Ticket Blotter Strategy: [] Test Account: T.CH7 | TEST

Stock Future Option MultiLeg

Order Type / Post Type Limit Class Exchanges Ref SDiv

ActiveTaker PostLimit Simple Market Exchanges (All) Locked Send New Update Existing

Auto Hedge Time In Force Limit Prc Adjuster (value / reference) Take Reach / Size Release / Progress Expose CxlOverlap / Max Visible / Exchanges / Public Size Opx Floor UPrc Range (%)

SR.AUTO Day 0.00 abs / cro +bid +ask Immediate Immediate 0 sec CxlOverlap 100 % all NumExch: 1 PublicSize: Default Opx Floor 0.00 UPrc Range (%) -2.00 - 2.00

	AAPL 16-SEP-2022	Side	Size	uLimitPrc	Open	Filled	State	Limit	Child Orders	CumBSz	oBid	oAsk	CumASz	iBid	iAsk	uBid	uAsk	sVol	sPrc	optVe
X	140.00 CALL	SELL	406		0	406	DONE	9.490	[none]	73	9.50	9.60	2.435	0.01	123.48	149.53	149.54	77.57	9.537	0.000
X	141.00 CALL	SELL	122		0	122	DONE	8.490	[none]	73	8.50	8.60	1.312	0.01	112.33	149.53	149.54	76.33	8.539	0.000
X	142.00 CALL	SELL	657		0	657	DONE	7.490	[none]	81	7.50	7.60	1.809	0.01	101.07	149.53	149.54	72.39	7.542	0.001
X	144.00 CALL	BUY	417		0	417	DONE	5.610	[none]	231	5.50	5.60	878	0.01	78.21	149.53	149.54	57.21	5.544	0.001
X	145.00 CALL	BUY	141		0	141	DONE	4.610	[none]	444	4.50	4.60	681	0.01	66.52	149.53	149.54	49.80	4.547	0.002

Vega Limits (Buy|Sell|Abs) L: none S: none A: none R: 1.00 Premium Limits (Buy|Sell|Abs) L: none S: none A: none Contract Limits (Buy|Sell|Abs) L: none S: none A: none Dollar Delta Limits (Buy|Sell) L: 20,741,958 S: 20,741,958 Auto AutoSize MktSize

Active Today History Staged

RiskGrp	Restore	Accont	Ticker(s)	Type	nSt	nOp	nFut	OrderTime	Cancel	State	OpnCn	OpnSh	OpnFut	FillCn	FillSh	FillFut	FillVe	FillPr	FillId	VeBot	VeSld	LastFillTm	LastUpdate	Date	UserName
4CF4	*	T.CH7	AAPL	OPT	2	6	0	13:04:24		ACTIVE	0	70,332	0	2,221	30,395	0	136	3,000	-2,949	208	72	13:04:32	13:04:32	<today>	felasha long

Order Risk Group Limits

SpiderRock execution engines maintain automated risk counters that keep track of net risk as child orders fill.

These order risk group limits are set by the user using the limit fields at the bottom of the ticket (see red box below) on individual basket orders. These risk group limits act to control the available risk by limiting the child order size.

The risk group limits can be expressed as follows:

Long Vega
Short Vega
Absolute Vega
Vega Ratio

Long Premium
Short Premium
Absolute Premium

Long number of Contracts
Short number of Contracts
Absolute number of Contracts

The long (L), short (S), and absolute (A) fields can be used to size each parent order by clicking on the AutoSize button to the right of these limit boxes which will populate the size corresponding to the risk limits.

Spider Ticket Blotter (v7.0) © 2022 SpiderRock Technology, LLC (user: luke.walker **SRAdmin**) (2022-09-26 14:33:32.853)

SPIDERROCK Ticket Blotter Strategy: test 1: luke.walker Test Account: TLW1

Stock Future Option MultiLeg

Order Type / Post Type Limit Class Exchanges Ref SDiv

ActiveTaker PostLimit Simple Prc Exchanges (All) Locked Send New Update Existing

Auto Hedge Time In Force Limit Prc Adjuster (value / reference) Take Reach / Size Release / Progress Expose CxlOverlap / Max Visible / Exchanges / Public Size Opx Floor UPrc Range (%)

NONE Day 0.00 abs / surf +bid +ask Immediate Immediate 0 sec CxlOverlap 100 % all NumExch: 1 PublicSize: Default Opx Floor 0.00 UPrc Range (%) -2.00 - 2.00

	AAPL 16-DEC-2022	Side	Size	uLimitPrc	opt.position	ssale.flag	Open	Filled	State	Limit	Child Orders	iBid	oBid	oAsk	iAsk	CumBSz	CumASz	sPrc	sVol
X	145.00 CALL	BUY	186	14.75	OPN	SH	0	0	NONE			37.91	14.75	14.85	38.28	2,593	814	14.803	38.10
X	150.00 CALL	BUY	177	11.75	OPN	SH	0	0	NONE			36.56	11.70	11.80	36.91	2,074	1,326	11.751	36.74
X	155.00 CALL	BUY	174	9.05	OPN	SH	0	0	NONE			35.20	9.00	9.10	35.55	8,867	1,690	9.064	35.42

	SPY 16-DEC-2022	Side	Size	uLimitPrc	opt.position	ssale.flag	Open	Filled	State	Limit	Child Orders	iBid	oBid	oAsk	iAsk	CumBSz	CumASz	sPrc	sVol
X	364.00 CALL	SELL	73	20.59	OPN	SH	0	0	NONE			27.02	20.63	20.76	27.21	328	985	20.693	27.11
X	365.00 CALL	SELL	72	20.00	OPN	SH	0	0	NONE			26.91	20.05	20.18	27.09	148	1,193	20.109	26.99
X	366.00 CALL	SELL	72	19.43	OPN	SH	0	0	NONE			26.77	19.47	19.60	26.96	356	947	19.541	26.87

Vega Limits (Buy|Sell|Abs) L: 5,000 S: 5,000 A: none R: 1.00 Premium Limits (Buy|Sell|Abs) L: none S: none A: none Contract Limits (Buy|Sell|Abs) L: none S: none A: none Dollar Delta Limits (Buy|Sell) L: 31,072,944 S: 31,072,944 Auto AutoSize MktSize

AAPL / 365D <none> / 61A7



The example above displays a ticket for a basket comprised of 3 sell orders in SPY and 3 buy orders in AAPL. Each strike on both the buy and sell sides were sized using a 5,000 vega limit. The user input 5,000 in the L: (long) box and 5,000 vega in the S: (short) box and clicked on AutoSize (see red arrow). This converts the vega limits to the appropriate number of contracts on each line. The down arrow next to the AutoSize button reduces the size by 20% on each click. Reducing the size on each line may allow for a better distribution of fills across more strikes.

The MktSize button allows user to load the current public market size on each strike of the basket.

Spider Ticket Blotter (v7.0) © 2022 SpiderRock Technology, LLC (user: luke.walker **SRAdmin**) (2022-09-26 14:33:32.853)

SPIDERROCK Ticket Blotter

Strategy: test1:luke.walker Test Account: TLW1

Stock Future Option MultiLeg

Order Type / Post Type: ActiveTaker PostLimit Limit Class: Simple Prc Exchanges: Exchanges (All) Ref SDiv: Locked Send New Update Existing

Auto Hedge: NONE Time In Force: Day Limit Prc Adjuster (value / reference): 0.00 abs surf +bid +ask Take Reach / Size Release / Progress Expose: Immediate Immediate 0 sec CxlOverlap / Max Visible / Exchanges / Public Size: CxlOverlap 100 % all NumExch: 1 PublicSize: Default Opx Floor: 0.00 UPrc Range (%): -2.00 2.00

AAPL 16-DEC-2022										Side	Size	uLimitPrc	opt.position	ssale.flag	Open	Filled	State	Limit	Child Orders	iBid	oBid	oAsk	iAsk	CumBSz	CumASz	sPrc	sVol
X	145.00	CALL	BUY	186	14.75	OPN	SH	0	0	NONE										37.92	14.75	14.85	38.29	2,142	90	14.799	38.10
X	150.00	CALL	BUY	177	11.75	OPN	SH	0	0	NONE										36.57	11.70	11.80	36.92	3,211	262	11.744	36.72
X	155.00	CALL	BUY	174	9.05	OPN	SH	0	0	NONE										35.38	9.05	9.10	35.56	1,655	21	9.054	35.40

SPY 16-DEC-2022										Side	Size	uLimitPrc	opt.position	ssale.flag	Open	Filled	State	Limit	Child Orders	iBid	oBid	oAsk	iAsk	CumBSz	CumASz	sPrc	sVol
X	364.00	CALL	SELL	73	20.59	OPN	SH	0	0	NONE										27.10	20.73	20.85	27.27	236	16	20.751	27.13
X	365.00	CALL	SELL	72	20.00	OPN	SH	0	0	NONE										26.98	20.14	20.27	27.16	184	16	20.166	27.01
X	366.00	CALL	SELL	72	19.43	OPN	SH	0	0	NONE										26.84	19.56	19.69	27.03	1,795	16	19.597	26.89

Vega Limits (Buy|Sell|Abs): L: 1,000 S: 1,000 A: 10,000 R: 1.00 Premium Limits (Buy|Sell|Abs): L: none S: none A: none Contract Limits (Buy|Sell|Abs): L: none S: none A: none Dollar Delta Limits (Buy|Sell): L: 31,072,944 S: 31,072,944 Auto AutoSize MktSize

AAPL / 365D <none> / 61A7

In addition, the user can implement group controls for the entire basket. By changing the long limit from 5,000 to 1,000 and changing the short limit from 5,000 to 1,000 we can limit the leg exposure to 1,000 vega on either side. Furthermore, by implementing a 10,000 vega limit in the absolute vega box, this will limit the basket's absolute vega risk to 10,000 vega. For more details and additional examples, please see Parent Order Risk Controls document.

Dollar delta limits (see green arrow) are used to control the dollar delta exposure of both the options and any autohedge. SpiderRock uses a trailing 30 day average compared to the current day's volume to calculate how much dollar delta an underlying can absorb in a hedging period. The dollar delta exposure is restricted in order to minimize and control the market impact of the autohedge.



MultiLeg Ticket

- MultiLeg Characteristics:

The SpiderRock Multi-leg orders can include up to 6 option legs and 1 stock leg. At minimum, 2 legs are required.

Spread limit price can be static or tied to stock.

Option fills can be automatically hedged with corresponding stock.

- Order Type

Seeker (Atomic Execution) - Will send child orders to multiple exchange complex-order-books (COB). When a fill occurs Seeker will rebalance the remaining size to the exchange that is trading. If the limit price is marketable, Seeker will remove the liquidity.

COB Maker (Atomic Execution) - Will post child orders to multiple exchange complex-order-books (COB). Allows the client to leverage any of the SpiderRock posting algos (see page 8). COB maker orders will never remove liquidity.

The following order types introduce leg risk that is assumed by the user. These order types will post orders at a price level that will allow our system to complete the multileg order by sending immediate or cancels (IOC). Our system will also restrict the posted size based on the amount of available liquidity on the screen to complete the order.

Seek/Leg (Hybrid Atomic/Non-Atomic Execution) - Will work a multileg order flashing the exchange complex-order-books (COB) for 5 seconds. If the order is not filled or is only partially filled in the COBs, the engine will then separately work each leg of the order on regular exchange order books.



Legger (Non-Atomic Execution) - Will work each leg separately on regular exchange order books.

- **Mitigate Leg Risk for Non-Atomic Execution**

The user can set Exposure, Slippage, and Lead Side to mitigate leg risk. Exposure limits the size to a maximum size amount (based on a % of total order size). Slippage represents the worst amount that a user is willing to pay in order to complete a legged package. The slippage amount (in pennies) only comes into play if the order is legged. Being legged means that part of the order (the posted leg) gets filled and our system missed the completion of the other leg when sending an IOC. This can occur when the market is moving fast. Slippage gives room to chase a single leg if needed. Once that leg fills and the package unit is whole, the order continues to work at the original order price. Lead Side allows the user to set which side of the order will be posting and thus filling first.

With any of our multi-leg strategies, the user can choose a posting algo to work a multi-leg order. If the user does not select a posting strategy and is using Seeker/Leg or Legger then the order will only take liquidity when both legs are available at the user price. The SpiderRock Probabilities and Volatility Surface can also be leveraged as secondary limits.

Pair Trade Ticket

This order type has inherent leg risk assumed by the user. The SpiderRock engine will post the lead leg at a level that allows the engine to complete the pair at the desired package price by removing liquidity.

To mitigate leg risk, the user can use Exposure and Slippage.

- Exposure restricts the posted size to a maximum size amount based on a % of total order size.
- Slippage gives room to complete the package via a more aggressive price up to a pre-defined margin greater than the original order price. The slippage amount (in pennies) only comes into play if the order is legged. Being legged means that the posted leg gets filled and our engine missed the completion of the other legs while removing liquidity. Once the legged package fills and the package unit is whole, the order continues to work at the original order price. Further, the SpiderRock engine will also post size based on the amount of take size available on the other legs.

Functionalities

Spider Pair Trade (v7.0) © 2020 SpiderRock Technology, LLC (user: luke.walker **SRAdmin**) (2022-09-07 14:24:48.772)

SPIDERROCK Pair Trade

Strategy: Test Account: TLWL

123

456

Post Type

Order Size

Order Price

Limit Type

Auto Hedge

Time In Force

Exposure / Slippage

UPrc Range (%)

New Active

New Held

Update C66E

PostLimit

10

0.37

debt

mid

surf

Prc

NONE

Day

5%

0.00

-2.00

2.00

Side	Ratio	Lead	Instrument	PosMark	uLimitPrc	effLimitPrc	Open	Filled	State	Limit	Child Orders	UPrc	BidSz	iBid	BidPrc	AskPrc	iAsk	AskSz	sVol	sPrc	Vega
X SELL	1	☒	AMZN JAN-20 '23: 130 C		13.021	13.050	1	1	WORK		+ [1 @ 13.05]	129.33	2,209	39.63	12.90	13.00	39.95	470	39.80	12.95	-31
X BUY	1	☒	AAPL JAN-20 '23: 155 C		13.391	13.350	0	1	HOLD		[none]	156.11	4,816	31.70	13.40	13.55	32.11	3,841	31.91	13.48	37
Total:					0.370	0.300								100.00	0.40	0.65	100.00		100.00	0.52	6

1. Users can choose a posting algo to work a pair order (PostWith, PostWthF, PostTurn, PostImpr, PostLimit, PostFlash, PostFlashI, PostFlashW). Please refer to page 9 for posting order description.

SPIDERROCK Pair Trade

1 2 3		Post Type		Order Size		Order Price		Limit Type			
4 5 6		PostLimit		10		0.55		Prc			
Side		Ratio		Lead		Instrument		PosMark		uLimitPrc	
X SELL		1		✓		AMZN JAN-20 '23: 130 C		13.112		13.112	
X BUY		1		✓		AAPL JAN-20 '23: 155 C		13.662		13.662	
Total:								0.550			



2. Users can select a lead leg by checking the box to the right of the ratio. The lead legs are the legs that will post in the market first. As soon as the lead legs are filled, the non- posting legs would take liquidity to complete the package.

Important note: If all lead legs or no lead legs are checked when using a posting strategy, then all legs can post.

PT Spider Pair Trade (v7.0) © 2020 SpiderRock Technology, LLC (user: luke.walker)

SPIDERROCK Pair Trade

1 2 3 Post Type Order Size Order Price
4 5 6 PostLimit 10 0.55 debit mid surf

	Side	Ratio	Lead	Instrument	PosMark
x	SELL	1	☒	AMZN JAN-20 '23: 130 C	
x	BUY	1	☐	AAPL JAN-20 '23: 155 C	
Total:					

3. New Active: creates a new order that is immediately working. Order can be paused using HOLD from the lower grid.

New Held: creates a new order that is paused on creation. This order will need to be released using RELEASE from the lower grid.

Update (risk groupID): Allows for an order algo parameters to be modified, including size, order price, posting logic, etc. The risk groupID of the modified order appears under the Update box as displayed below (example: C66E) Modifying the order price can also be done using the LIMIT+ (increasing price), LIMIT- (decreasing price) or JOIN (joining the NBBO price).

Upper Grid

New ActiveNew HeldUpdate C66E						
AskPrc	iAsk	AskSz	sVol	sPrc	Vega	Delta
10.0	40.13	2,705	39.82	13.00	-31.0	-0.563
10.0	32.19	5,434	31.90	13.49	37.1	0.578
10.5	100.00		100.00	0.49	6.1	0.015

Lower Grid

Strategy:Test Account: TLWL

Exposure / SlippageUPrc Range (%)New ActiveNew HeldUpdate C66E

Child Orders	UPrc	BidSz	iBid	BidPrc	AskPrc	iAsk	AskSz	sVol	sPrc	Vega
+ [1 @ 13.25]	129.63	503	39.73	13.10	13.20	40.05	1,849	39.78	13.12	-31.0
[none]	156.61	4,078	31.73	13.70	13.85	32.13	5,084	31.91	13.77	37.1
			100.00	0.50	0.75	100.00		100.00	0.65	6.1

RELEASEHOLDLIMIT +LIMIT -JOIN

TODAY'S STRATEGIES

PairBidSz	PairBidPx	PairAskPx	PairAskSz	Strategy	LastUpdate
1,849	0.50	0.75	503	like.walker	14:29:01



SYMBOL VIEWER HOT KEYS

Hot Key	Result
F5	Refresh symbol data
F6	Create closing order using position column
F7	Create order using test column
Esc	Clear all test positions from columns
Ctrl+~	Scroll through user account list
Ctrl+T	Move focus to the ticker on the main SV page
Ctrl+B	Opens the ticket blotter
Ctrl+B (in a quick ticket)	Will buy using the parameters in the quick ticket
Ctrl+S (in a quick ticket)	Will sell using the parameters in the quick ticket
Select option prices and hold Shift and hit any number from 1 to 6	Loads the order parameters corresponding to the saved preset number in a multi-leg ticket
Select option prices and hold Ctrl and hit any number from 1 to 6	Loads the order parameters corresponding to the saved preset number in an options ticket
Shift+up/down (in a quick ticket)	Increments the price up or down by 10 cents
PgUp/PgDn (in a quick ticket)	Increments the price up or down by 5 cents
Ctrl+Autosize button (in the ticket blotter)	The last entry in the risk limit fields are cleared
Hold Shift + hit restore button (in the ticket blotter)	Changes the symbol in the main Symbol Viewer page to view the symbol on that order

